

CURRICULUM VITAE

(September, 2026)

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Personal data

Name

Juan Francisco Spedaletti

Place and date of birth

San Luis (capital), Argentina, 28/12/1980

Citizenship

Argentine

Telephone

266-4652202

Marital status

Single

Children

None

National ID (DNI)

27932507

Passport

AAA871769

Tax ID (CUIL)

23-27932507-9

E-mail

jfspedaletti@unsl.edu.ar or jfspedaletti@gmail.com

ResearchGate profile

https://www.researchgate.net/profile/Juan_Spedaletti2

ORCID

<https://orcid.org/0009-0007-4423-1758>

Personal webpage

<https://juanspedaletti.github.io>

Professional address

Address

Instituto de Matemática Aplicada San Luis (IMASL)-CONICET, Avenida Italia 1556, first floor, office 155. San Luis, Argentina

Telephone

+54(0266)4520300, extension: 3053

Education and training

Secondary education

Bachiller (secondary school diploma), Colegio Juan Crisostomo Lafinur (1995-2000). San Luis (capital), Argentina.

University education

1. Licenciatura in Computer Science (2000-2002, **not completed**). Universidad Nacional de San Luis (UNSL).
2. Licenciado in Mathematical Sciences (**5-year programme**, 2002-2008). Universidad Nacional de San Luis (UNSL).

Diploma link

3. Master of Science in Mathematics (2011-2014). Universidad Nacional de San Luis (UNSL). **Thesis grade: outstanding (highest grade).**

Diploma link (front)

Diploma link (back)

4. Ph.D. in Mathematical Sciences (2011-2016). Universidad Nacional de San Luis (UNSL). **Thesis grade: outstanding (highest grade).**

Diploma link (front)

Diploma link (back)

Current positions

1. Assistant Professor (Profesor Adjunto), full-time, tenured (**position currently on leave due to incompatibility of appointments**)
2. Associate Professor (Profesor Asociado), full-time, acting appointment.
3. Assistant Researcher, CONICET (since March 2021)

Link to CONICET resolution 633/2021

Research interests

Analysis

Calculus of variations

Ordinary differential equations

Nonlinear differential equations

Nonlocal operators

Shape optimization problems

Eigenvalue problems for problems with nonstandard growth

Graduate courses completed in Argentina

1. Theory of Monotone Operators. Venue: Departamento de Matemática, Universidad Nacional de San Luis. Instructor: Dr. Carlos Zuppa (Universidad Nacional de San Luis). Workload: 150 hs. Grade: 9 (nine). Date: 24/08/2010 to 24/12/2010.

Resolution link: RR-3-355/12 UNSL

2. Nonlinear Differential Equations. Venue: Departamento de Matemática, University of Buenos Aires. Instructor: Dr. Pablo De Napoli (University of Buenos Aires). Workload: 96 hs. Grade: 10 (ten). Date: May 2012.

Certificate link

3. Nonlinear Equations and Geometric Optics. Venue: Departamento de Matemática, University of Buenos Aires. Instructor: Dr. Cristian Gutierrez (Temple University). Workload: 24 hs. Grade: 10 (ten). Date: December 2012.

Certificate link

4. Applied Mathematics (Probability Theory). Venue: Instituto de Matemática Aplicada San Luis. Instructor: Dr. Juan Carlos Cesco (Universidad Nacional de San Luis). Workload: 80 hs. Grade: 10 (ten). Date: July 2013.

Resolution link: RD-3-777/14 UNSL

5. Seminar 1. Venue: Departamento de Matemática, University of Buenos Aires. Instructor: Dr. Julián Fernández Bonder (University of Buenos Aires). Workload: 80 hs. Grade: 10 (ten). Date: December 2012.

Resolution link: RD-3-642/13 UNSL

6. Research Seminar. Venue: Departamento de Matemática, University of Buenos Aires. Instructor: Dr. Julián Fernández Bonder (University of Buenos Aires). Workload: 80 hs. Grade: 10 (ten). Date: December 2012.

Resolution link: RD-3-642/13 UNSL

7. Nonlinear Differential Equations and Homogenization. Venue: Instituto de Matemática Aplicada San Luis. Instructor: Dr. Julián Fernández Bonder (University of Buenos Aires). Workload: 60 hs. Grade: 10 (ten). Date: March 2013.

Resolution link: RR-3-14/14 UNSL

8. Algebra. Venue: Departamento de Matemática, Universidad Nacional de San Luis. Instructor: Dr. Daniel Jaume (Universidad Nacional de San Luis). Workload: 150 hs. Grade: pass. Date: August 2013.

Resolution link: RD-3-755/13 UNSL

9. Analysis. Venue: Instituto de Matemática Aplicada San Luis (IMASL). Instructor: Dr. Felipe Zó (Universidad Nacional de San Luis). Grade: 9 (nine). Date: September 2014.

Resolution link: RD-3-643/14 UNSL

10. Epistemology. Instructor: Violeta Guyot. Venue: Facultad de Ciencias Humanas (Universidad Nacional de San Luis). Workload: 80 hs. Grade: 8 (eight). Date: November 2014.

Resolution link: RR-4-1943/15 UNSL

Graduate courses completed abroad

1. Regularity of the free boundary in problems with distributed sources. Instructor: Sandro Salsa (Politecnico di Milano). Venue and date: Universitat Politècnica de Catalunya, Barcelona, Spain, 15-20 July 2013.

Attendance certificate link

Event webpage

2. Regularity results for nonlocal equations. Instructor: Luis Silvestre (University of Chicago). Venue and date: Universitat Politècnica de Catalunya, Barcelona, Spain, 15-20 July 2013.

Attendance certificate link

Event webpage

3. Regularity results in free boundary problems. Instructor: Alessio Figalli (University of Texas at Austin). Venue and date: Universitat Politècnica de Catalunya, Barcelona, Spain, 16-20 June 2014.

Event webpage

4. Flow of nonsmooth vector fields and applications. Part I. Instructor: Luigi Ambrosio (Scuola Normale Superiore, Pisa, Italy). Venue and date: IMPA, Rio de Janeiro, 6-10 July 2015.

Certificate link**Event webpage**

5. Flow of nonsmooth vector fields and applications. Part II. Instructor: Alessio Figalli (University of Texas at Austin, Texas, USA). Venue and date: IMPA, Rio de Janeiro, 13-17 July 2015.

Certificate link**Event webpage**

6. Free Boundary Problems: Common aspects and methods. Instructor: Luis Caffarelli (University of Texas at Austin, Texas, USA). Venue and date: IMPA, Rio de Janeiro, 6-17 July 2015.

Certificate link**Event webpage****Other graduate courses attended as an auditor**

1. Finite Elements. Venue: Departamento de Matemática, University of Buenos Aires. Instructor: Gabriel Acosta Rodriguez (University of Buenos Aires).
2. Geometric Measure Theory. Venue: Departamento de Matemática, University of Buenos Aires. Instructor: Gabriel Acosta Rodriguez (University of Buenos Aires).
3. Γ -convergence and Applications. Instructor: Julián Fernández Bonder (University of Buenos Aires). Venue: Departamento de Matemática, University of Buenos Aires. Date: second semester 2015.

Fellowships

1. Type I graduate fellowship. Awarding institution: CONICET (2011-2014).

Resolution link

2. Type II graduate fellowship. Awarding institution: CONICET (2014-2016).

Resolution link

Research stays abroad

1. Departament de Matemàtiques, Universitat Politècnica de Catalunya. JISD workshops (Workshop on Interactions Between Dynamical Systems and Partial Differential Equations) 2013 and JISD 2014. 15-19 June 2013 and 16-20 June 2014.

Link

2. Departamento de Matemática y Ciencia de la Computación, Universidad de Santiago de Chile (USACH), 2-13 October 2017, Santiago de Chile, Chile. Research stay in the area of nonlocal elliptic partial differential equations.

Link

3. Departamento de Matemática, Universidad Técnica Federico Santa María (UTFSM), 16-31 October 2017, Valparaíso, Chile. Research stay in the area of nonlocal elliptic partial differential equations.

Link

Teaching and student supervision

FACULTY MEMBER AT THE DEPARTAMENTO DE MATEMÁTICA, UNIVERSIDAD NACIONAL DE SAN LUIS (UNSL), SINCE MAY 2005 (17 YEARS AND TEN MONTHS OF TEACHING SENIORITY).

CATEGORIZED AS TEACHING RESEARCHER: CATEGORY 5 (LAST CATEGORIZATION 2014)

Undergraduate teaching

1. (2005-2009) Teaching Assistant, second category:

Algebra I-Mathematics I (all degree programmes of the FCFMyN, Food Engineering), Algebra II (Licenciatura and Teaching Degree in Mathematics, Electrical Eng. with orientation in Digital Systems, Licenciatura and Teaching Degree in Computer Science, Licenciatura and Teaching Degree in Physics, Food Eng., Mining Eng., Computer Eng., Informatics Eng.) and Numerical Analysis (Licenciatura and Teaching Degree in Mathematics, Food Eng., Computer Eng., Informatics Eng., Mining Eng.).

Resolutions (my surname changed; the national ID number is the same): Link RDMat-3-20/05 UNSL,

Link RDMat-3-23/06 UNSL,

Link RD-3-248/07 UNSL

Link RDMat-3-28/07 UNSL,

Link RD-3-455/09 UNSL and

Link RD-3-813/09 UNSL

2. (2010-2012) Teaching Assistant, first category, part-time, temporary and substitute appointments

- First semester 2010: Mathematics I (Pharmacy, Licenciatura in Food Science and Technology, University Technical Degree in Sterilization, University Technical Degree in Biological Laboratory, Licenciatura in Biochemistry, Licenciatura in Chemistry, Teaching Degree in Chemistry, Chemical Analyst, Licenciatura in Biotechnology).
- Second semester 2010: Numerical Analysis (Licenciatura and Teaching Degree in Mathematics, Food Eng., Computer Eng., Informatics Eng., Mining Eng.).
- First semester 2011: study leave.
- Second semester 2011: Numerical Analysis (Licenciatura and Teaching Degree in Mathematics, Food Eng., Computer Eng., Informatics Eng., Mining Eng.).
- First semester 2012: Mathematics I (Pharmacy, Licenciatura in Food Science and Technology, University Technical Degree in Sterilization, University Technical Degree in Biological Laboratory, Licenciatura in Biochemistry, Licenciatura in Chemistry, Teaching Degree in Chemistry, Chemical Analyst, Licenciatura in Biotechnology).
- Second semester 2012: Numerical Analysis (Licenciatura and Teaching Degree in Mathematics, Food Eng., Computer Eng., Informatics Eng., Mining Eng.).

Resolutions:

Link RDMat-3-8/10 UNSL,

Link RDMat-3-22/11 UNSL and

Link RD-3-160/12 UNSL

3. (2013-2015) Teaching Assistant, first category, part-time, temporary and tenured appointments

- First semester 2013: Mathematics I (Pharmacy, Licenciatura in Food Science and Technology, University Technical Degree in Sterilization, University Technical Degree in Biological Laboratory, Licenciatura in Biochemistry, Licenciatura in Chemistry, Teaching Degree in Chemistry, Chemical Analyst, Licenciatura in Biotechnology).

- Second semester 2013: Numerical Analysis (Licenciatura and Teaching Degree in Mathematics, Food Eng., Computer Eng., Informatics Eng., Mining Eng.).
- First semester 2014: Mathematics and Computing II (Licenciatura in Mathematics and Licenciatura in Applied Mathematics).
- Second semester 2014: Equations of Mathematical Physics (Licenciatura in Mathematics and Licenciatura in Applied Mathematics).
- First semester 2015: Calculus I (Licenciatura and Teaching Degree in Computer Science).
- Second semester 2015: Equations of Mathematical Physics (Licenciatura in Mathematics and Licenciatura in Applied Mathematics).

Resolutions:

Link RCD-3-114/12 UNSL and

Link RCD-3-148/12 UNSL

4. (2016) Senior Teaching Assistant (Jefe de Trabajos Prácticos), interim:
First semester 2016: Mathematics and Computing I (Licenciatura in Mathematics and Licenciatura in Applied Mathematics).

Resolution:

Link RDMat-3-6/16 UNSL

5. (2016-2019) Assistant Professor (Profesor Adjunto), full-time, acting and tenured appointment:
 - Second semester 2016: Algebra I (Computer Eng., Informatics Eng., Mining Eng., Licenciatura in Computer Science, Teaching Degree in Electronics Technology, Teaching Degree in Mathematics, Licenciatura in Mathematics, Licenciatura in Applied Mathematics, Electrical Eng., Food Eng., Teaching Degree in Computer Science and Licenciatura in Geology).
 - First semester 2017: Algebra I (Computer Eng., Informatics Eng., Mining Eng., Food Eng., Licenciatura in Computer Science, Teaching Degree in Electronics Technology, Electrical Eng., Teaching Degree in Computer Science and Licenciatura in Geology).
 - Second semester 2017: Equations of Mathematical Physics (Licenciatura in Mathematics and Licenciatura in Applied Mathematics).
 - First semester 2018: Finite Elements I (Licenciatura in Applied Mathematics).

- Second semester 2018: Numerical Analysis (Informatics Eng., Computer Eng., Licenciatura in Mathematics and Licenciatura in Applied Mathematics).
- First semester 2019: Finite Elements I (Licenciatura in Applied Mathematics).
- Second semester 2019: Numerical Analysis (Informatics Eng., Computer Eng., Licenciatura in Mathematics and Licenciatura in Applied Mathematics).

Resolutions:

Link RDMat-3-51/16 UNSL,

Link RDMat-3-75/18 UNSL,

Link RDMat-3-117/18 UNSL,

Link RDMat-3-10/19 UNSL,

Link RDMat-3-41/19 UNSL and

Link RCS-3-193/19 UNSL

6. (2020-present) Associate Professor (Profesor Asociado), full-time, acting appointment:

- First semester 2020: Algebra I (Informatics Eng., Computer Eng. and Licenciatura in Computer Science).
- Second semester 2020: Equations of Mathematical Physics (Licenciatura in Mathematics and Licenciatura in Applied Mathematics).
- First semester 2021: Algebra I (Informatics Eng., Computer Eng. and Licenciatura in Computer Science).
- Second semester 2021: Algebra I (Informatics Eng., Computer Eng. and Teaching Degree in Computer Science).
- First semester 2022: Algebra I (Informatics Eng., Computer Eng. and Teaching Degree in Computer Science).
- Second semester 2022: Numerical Analysis (Informatics Eng., Computer Eng., Licenciatura in Mathematics and Licenciatura in Applied Mathematics).
- First semester 2023: Algebra I (Informatics Eng. and Computer Eng.).
- First semester 2024: Mathematics and Computing II (Licenciatura in Mathematics).
- Second semester 2024: Numerical Analysis (Informatics Eng., Computer Eng., Licenciatura in Mathematics and Licenciatura in Applied Mathematics).

- First semester 2025: Mathematics and Computing II (Licenciatura in Mathematics).
- Second semester 2025: Numerical Analysis (Informatics Eng., Computer Eng., Licenciatura in Mathematics and Licenciatura in Applied Mathematics).

Resolutions:

Link RCD-3-12/20 UNSL,

Link RDMat-3-2/21 UNSL and

Link RDMat-3-107/21 UNSL

Teaching in the entrance course of the Facultad de Ciencias Físico Matemáticas y Naturales, Universidad Nacional de San Luis

1. (2008) Group assistant. **Resolution:** Link RD-3-74/08 UNSL
2. (2012) Group leader. **Resolution:** Link RD-3-104/12 UNSL
3. (2013) Group leader. **Resolution:** Link RD-3-137/13 UNSL
4. (2023) Group leader. **Resolution:** IN PROGRESS

Elective undergraduate and graduate courses taught

1. Nonlocal Operators and Fractional Calculus. Elective course taught for: Master's programme in Mathematics. Role: instructor in charge (theory and problem sessions). First semester 2018.

Resolution: Link RD-3-468/18 UNSL

2. Nonlocal Operators and Optimal Design Problems. Elective course taught for: Master's programme in Mathematics. Role: instructor in charge (theory and problem sessions). First semester 2019.

Resolution: Link RD-3-784/19 UNSL

3. Introduction to Orlicz-Sobolev Spaces and Applications. Elective course for the Licenciatura in Mathematics and the Master's programme in Mathematics. First semester 2022.

Resolutions:

Link RD-3-118/22 UNSL and

Link RD-3-129/22 UNSL

Organizer of visiting professors' stays (scientific collaboration, courses and talks)

1. Visit of Dr. Julián Fernández Bonder (University of Buenos Aires-CONICET), who taught the course “Nonlinear Differential Equations” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: second semester 2013. Funded by the inter U programme.

Resolution link RR-3-14/14 UNSL

2. Visit of Dr. Julián Fernández Bonder (Departamento de Matemática, University of Buenos Aires-CONICET), who taught the course “Non-local Operators and Fractional Calculus” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: 6-10 March 2017. Funded by the Departamento de Matemática, Universidad Nacional de San Luis.

Resolution link: RDMat-3-68/16

3. Visit of Dr. Ariel Salort (University of Buenos Aires-CONICET), who taught the course “The Fučík Spectrum” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: 6-10 November 2017. Funded by the Departamento de Matemática, Universidad Nacional de San Luis.

Resolution link: RDMat-3-23/17

4. Visit of Ph.D. João Vitor da Silva (Universidade de Brasília), who taught the course “Regularity issues in elliptic equations and free boundary problems” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: 5-16 March 2018. Funded by the Departamento de Matemática, Universidad Nacional de San Luis.

Resolution link: RDMat-3-68/17

5. Visit of Dr. Erwin Topp Paredes (Departamento de Matemática y Ciencia de la Computación, Universidad de Santiago de Chile, US-ACH), who taught the course “Ergodic problems in Hamilton-Jacobi equations” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: 22-26 April 2019. Funded by the Departamento de Matemática and the Secretaría de Ciencia y Técnica, Universidad Nacional de San Luis.

Resolution link: RDMat-3-27/19

6. Visit of Dr. Julián Fernández Bonder (Departamento de Matemática, University of Buenos Aires-CONICET), who gave the talk “Fractional elliptic operators and optimal design problems” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: 22-26 July 2019. Funded by the Departamento de Matemática, Universidad Nacional de San Luis.

Resolution link: RDMat-3-61/19

7. Visit of Dr. Leandro Martin Del Pezzo (University of Buenos Aires-CONICET), who taught the minicourse “Harmonic functions on trees” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: 25-29 November 2019.

Resolution link: RDMat-3-118/19

8. Visit of Dr. Julián Fernández Bonder (Departamento de Matemática, University of Buenos Aires-CONICET), who taught the minicourse “Orlicz, Orlicz-Sobolev and fractional Orlicz-Sobolev spaces” and carried out scientific collaboration. Venue: Universidad Nacional de San Luis-Instituto de Matemática Aplicada San Luis. Period: 7-11 March 2022. Funded by the Departamento de Matemática, Universidad Nacional de San Luis.

Resolution link: RDMat-3-117/21

Student supervision

Undergraduate (Licenciatura) students

1. Licenciatura thesis in Mathematics. Mikhail Rios Guzman, Facultad de Ciencias Físico Matemáticas y Naturales, Universidad Nacional de San Luis. *Existence of solutions for the Dirichlet problem associated with the fractional Laplacian $(-\Delta_{p,\Omega})^s$* . **(NOT COMPLETED)**
2. Licenciatura thesis in Mathematics. Martín Sebastián Guzmán, Facultad de Ciencias Físico Matemáticas y Naturales, Universidad Nacional de San Luis. *Characterization of solutions for the Dirichlet problem associated with the Δ_ϕ -Laplacian operator*. (September 2021-November 2022, defended)

Link to thesis defence record

Master’s students

Master’s thesis in Mathematics. Natalí Ailín Cantizano, Facultad de Ciencias Físico Matemáticas y Naturales, Universidad Nacional de San Luis. *Conti-*

nuity of solutions for the Δ_φ -Laplacian operator. March 2019-present.

Resolution: Link RD-3-400/19 UNSL

Ph.D. students

Ph.D. thesis in Mathematics. Martín Sebastián Guzmán, Facultad de Ciencias Físico Matemáticas y Naturales, Universidad Nacional de San Luis. *Optimal design problems for state equations with nonstandard growth.* January 2023-present.

Resolution: Link RD-3-497/23 UNSL

Research activity and scientific output

Research projects (past)

I no longer participate in any of the research projects listed below. My research is currently self-funded through my own private investments, rather than through publicly funded research projects.

1. Former principal investigator of the project “Differential Equations” (PROICO 03-2023), Universidad Nacional de San Luis (UNSL). **Resolution:** Resolution link

2. Former principal investigator of the project “Differential Equations” (PROIPRO 03-2420), Universidad Nacional de San Luis (UNSL).

Resolution: Resolution link CS N° 189 UNSL

3. Former member of the project “Problems with Nonstandard Growth”(PICT, ANPCyT 2019-2019-03837). Principal investigator: Analía Silva.

Resolution link IF-2021-11960060-APN-FONCYT

4. Former member of the project PIP-CONICET, code KE4 11220210100238CO “Elliptic problems with nonstandard growth”. Principal investigator: Analía Silva.

Link to list of approved projects

Link to evaluation report

5. “Differential Equations and Stochastic Processes”. Principal investigator: Julián Fernández Bonder (University of Buenos Aires). Code: PIP 2009 845/10

6. Former member of the “Research Group on Partial Differential Equations”, (University of Buenos Aires). Head: Noemí Wolanski.

Link to the group webpage

7. Former member of the project “Real and Functional Analysis. Differential Equations ”(PROICO 03-0720). Principal investigator: Sergio Favier (Universidad Nacional de San Luis), CONICET-UNSL.

Resolution: Link CS N° 189 UNSL

8. Former co-principal investigator of the project “Differential Equations” (2018/2020), Universidad Nacional de San Luis (UNSL), code: 03-2418. Principal investigator: Analía Silva. Co-principal investigator: Juan Spedaletti.

Resolution: Link CS N° 126 UNSL

Articles published in peer-reviewed journals

1. (With J. Fernández Bonder). *A shape optimization problem for Steklov eigenvalues in oscillating domains*. ESAIM Control Optim. Calc. Var., **23** (2017), 373-390.

DOI: <https://doi.org/10.1051/cocv/2015050>

Scimago Journal & Country Rank index: 1,249-Q1 (2016)

ArXiv link

2. (With C. Baroncini and J. Fernández Bonder). *Continuity results with respect to domain perturbation for the fractional p -Laplacian*. Appl. Math. Lett., **75** (2018), 59-67.

DOI: <https://doi.org/10.1016/j.aml.2017.06.005>

Scimago Journal & Country Rank index: 1,457-Q1 (2017)

ArXiv link

3. (With J. Fernández Bonder). *Some nonlocal optimal design problems*. J. Math. Anal. Appl., **459** (2018), no. 2, 906–931.

DOI: <https://doi.org/10.1016/j.jmaa.2017.11.015>

Scimago Journal & Country Rank index: 1,103-Q2 (2016)

ArXiv link

4. (With J. Fernández Bonder and J. Rossi). *Optimal design problems for the first p -fractional eigenvalue with mixed boundary conditions*. Adv. Nonlinear Stud., **18** (2018), no. 2, 323–335.

DOI: <https://doi.org/10.1515/ans-2018-0001>

Scimago Journal & Country Rank index: 1,588-Q1 (2017)

ArXiv link

5. (With J. Fernández Bonder and A. Silva). *Uniqueness of minimal energy solutions for a semilinear problem involving the fractional Laplacian*. Proceedings of the American Mathematical Society, **147** (2019), no. 7, 2925–2936.

DOI: <https://doi.org/10.1090/proc/14530>

Scimago Journal & Country Rank index: 1,108-Q1 (2018)

ArXiv link

6. (With J. V. da Silva, A. M. Salort and A. Silva). *A constrained shape optimization problem in Orlicz-Sobolev spaces*. Journal of Differential Equations, **267** (2019), no. 9, 5493–5520.

DOI: <https://doi.org/10.1016/j.jde.2019.05.038>

Scimago Journal & Country Rank index: 2,525-Q1 (2018)

ArXiv link

7. (With N. A. Cantizano and A. M. Salort). *Continuity of solutions for the Δ_ϕ -Laplacian operator*. Proceedings of the Royal Society of Edinburgh Section A: Mathematics, **151** (2020), no. 4, 1355–1382.

DOI: <https://doi.org/10.1017/prm.2020.63>

Scimago Journal & Country Rank index: 1,080-Q1 (2019)

ArXiv link

8. (With J. Fernández Bonder and A. Silva). *Gamma convergence and asymptotic behavior for eigenvalues of nonlocal problems*. Discrete and Continuous Dynamical Systems, **41** (2021), no. 5, 2125–2140.

DOI: <https://www.aims sciences.org/article/doi/10.3934/dcds.2020355>

Scimago Journal & Country Rank index: 1,289-Q1 (2020)

ArXiv link

9. (With J. Fernández Bonder). *Ljusternik-Schnirelmann eigenvalues for the fractional m -Laplacian without the Δ_2 -condition*. Communications on Pure and Applied Analysis, **24** (2025), no. 6, 1094–1117.

DOI: <https://www.aims sciences.org/article/doi/10.3934/cpaa.2025025> **Scimago Journal & Country Rank index:** 0,806-Q2 (2023)

ArXiv link

Submitted articles

(With J. Fernández Bonder and M. S. Guzmán). *An eigenvalue problem for a nonlocal quasilinear anisotropic equation in fractional Orlicz Sobolev spaces without the Δ_2 -condition*. ArXiv preprint (2026), 18pp.

ArXiv link

Conference proceedings

III MACI, Bahía Blanca. Venue: Departamento de Matemática, Universidad Nacional del Sur, May 2011. "A quasi-Kačanov iterative method for quasi-linear problems", authors: Juan Spedaletti (Universidad Nacional de San Luis), Carlos Zuppa (Universidad Nacional de San Luis). MACI, 3(2011), 161-164

Link to proceedings

AND THE PROCEEDINGS OF THE CONFERENCES AT WHICH I
HAVE GIVEN COMMUNICATIONS

Theses

1. *An optimal design problem associated with the first Steklov eigenvalue* (Master's thesis in mathematics). Grade obtained: outstanding (highest grade).

Link to thesis

Resulting publication: *A shape optimization problem for Steklov eigenvalues in oscillating domains*. ESAIM Control Optim. Calc. Var., **23** (2017), 373-390.

DOI: <https://doi.org/10.1051/cocv/2015050>

2. *Asymptotic behaviour of optimal design problems for local and nonlocal models* (Ph.D. thesis in mathematics). Grade: outstanding (highest grade).

Link to thesis

Resulting publication: *Some nonlocal optimal design problems*. J. Math. Anal. Appl., **459** (2018), no. 2, 906–931.

DOI: <https://doi.org/10.1016/j.jmaa.2017.11.015>

Conference attendance

1. Meeting of the Unión Matemática Argentina (UMA). Venue: Facultad de Economía y Administración, Universidad Nacional del Comahue (2004), Neuquén, Argentina.
2. Attendance at the lecture by Dr. Luis Caffarelli, "Surface homogenization problems". Venue: Sociedad Científica Argentina, 26 July 2012, Buenos Aires, Argentina.

Event link

3. 1° Workshop on Analysis, Partial Differential Equations and Applied Mathematics. Venue: Instituto Argentino de Matemática (IAM-CONICET), 28-29 June 2012, Buenos Aires, Argentina.

Event link

4. Eleventh Workshop on Interactions Between Dynamical Systems And Partial Differential Equations (JISD2013). Venue: Universitat Politècnica de Catalunya, Barcelona, Spain, 15-20 July 2013.

Certificate link

5. Annual meeting of the Unión Matemática Argentina (UMA). Venue: Facultad de Ciencias Exactas, Ingeniería y Agrimensura, Rosario, Argentina, September 2013.

Link to conference communication

6. Eleventh Workshop on Interactions Between Dynamical Systems And Partial Differential Equations (JISD2014). Venue: Universitat Politècnica de Catalunya, Barcelona, Spain, June 2014.

Link to conference confirmation e-mail

7. Annual meeting of the Unión Matemática Argentina (UMA). Venue: Facultad de Ciencias Exactas y Naturales, Universidad Nacional de San Luis, San Luis, Argentina, September 2014.

Certificate link

8. X Americas Conference on Differential Equations and Nonlinear Analysis. Venue: Facultad de Ciencias Económicas (FCE), University of Buenos Aires (UBA), February 2015.

Link to poster presentation at the conference

9. International Workshop on Elliptic and Kinetic Differential Equations. Venue: IMPA, Rio de Janeiro, Brazil, 6-17 July 2015.

Certificate link

10. XIII National Meeting of Analysts A. P. Calderón. Venue: Hotel Edelweiss, Villa General Belgrano, Córdoba, 15-16 April 2016.

Link to conference participation

11. CAPDE School on Nonlinear Partial Differential Equations. Venue: Universidad Técnica Federico Santa María, Valparaíso, Chile, 28 September-2 October 2016, and Universidad de Santiago de Chile, 3-5 October 2016.

Link to e-mail from the conference organizers

12. Workshop on Partial Differential Equations and Applications. Venue: FAMAF, Universidad Nacional de Córdoba, Córdoba, Argentina, 27-29 September 2017.

Certificate link

13. Annual meeting of the Unión Matemática Argentina (UMA). Venue: Facultad de Ciencias Exactas y Naturales, University of Buenos Aires, Buenos Aires, Argentina, December 2017.

Certificate link

14. VII Congress on Applied, Computational and Industrial Mathematics. Venue: Departamento de Matemática, Universidad Nacional de Río Cuarto, Río Cuarto, Argentina, 8-10 May 2019.

Certificate link

15. Annual meeting of the Unión Matemática Argentina (UMA), virtUMA2020 (online).

Certificate link

16. X Congress on Applied, Computational and Industrial Mathematics. Venue: Universidad Tecnológica Nacional, Facultad Regional Córdoba, city of Córdoba, 12-15 May 2025.

Scientific communications, talks and posters

1. Annual meeting of the Unión Matemática Argentina (UMA). Venue: Facultad de Ciencias Exactas, Ingeniería y Agrimensura, Rosario, Argentina, September 2013. Communication presented: “Best Sobolev trace constant in oscillating domains”, authors: Julián Fernández Bonder (University of Buenos Aires), Juan Spedaletti (Universidad Nacional de San Luis).

Link to abstract

2. Eleventh Workshop on Interactions Between Dynamical Systems And Partial Differential Equations (JISD2014). Venue: Universitat Politècnica de Catalunya, Barcelona, Spain, June 2014. Poster presentation: “Asymptotic behaviour of the best Sobolev trace constant”, authors: Julián Fernández Bonder (University of Buenos Aires), Juan Spedaletti (Universidad Nacional de San Luis).

Link to poster presentation

3. Annual meeting of the Unión Matemática Argentina (UMA). Venue: Facultad de Ciencias Exactas y Naturales, Universidad Nacional de San Luis, San Luis, Argentina, September 2014. Communication presented: “An optimal expression for the shape derivative of the first eigenvalue of the Steklov problem”, authors: Julián Fernández Bonder (University of Buenos Aires), Juan Spedaletti (Universidad Nacional de San Luis).

Certificate link

4. X Americas Conference on Differential Equations and Nonlinear Analysis. Venue: Facultad de Ciencias Económicas (FCE), University of Buenos Aires (UBA), February 2015. Poster presentation: “Convergence results of the best Sobolev trace constant and optimal windows in oscillating domains”, authors: Julián Fernández Bonder (University of Buenos Aires), Juan Spedaletti (Universidad Nacional de San Luis).

Link to poster session

5. International Workshop on Elliptic and Kinetic Differential Equations. Venue: IMPA, Rio de Janeiro, Brazil, 6-17 July 2015. Communication presented: “Convergence results of the first Steklov eigenvalue and optimal windows in oscillating domains”, authors: Julián Fernández Bonder (University of Buenos Aires), Juan Spedaletti (Universidad Nacional de San Luis).

Certificate link

6. XIII National Meeting of Analysts A. P. Calderón. Venue: Hotel Edelweiss, Villa General Belgrano, Córdoba, 15-16 April 2016. Poster presentation: “Asymptotic behaviour for nonlocal problems”, authors: Julián Fernández Bonder (University of Buenos Aires), Juan Spedaletti (Universidad Nacional de San Luis).

Link to poster session

7. Seminar on Differential Equations and Numerical Analysis (SEDAN), Departamento de Matemática, University of Buenos Aires, August 2017. Communication presented: “Optimal design problems for the first p -fractional eigenvalue with mixed boundary conditions”, author: Juan Spedaletti (Universidad Nacional de San Luis).

Link to the seminar

8. Workshop on Partial Differential Equations and Applications. Venue: FAMAF, Universidad Nacional de Córdoba, Córdoba, Argentina, 27-29 September 2017. Communication presented: “Continuity results with respect to domain perturbation for the fractional p -Laplacian”, authors: Carla Baroncini (University of Buenos Aires), Julián Fernández Bonder (University of Buenos Aires) and Juan Spedaletti (Universidad Nacional de San Luis).

Certificate link

9. Seminar on Differential Equations, USACH. Venue: Departamento de Matemática y Ciencia de la Computación, Universidad de Santiago de Chile (Santiago, Chile), October 2017. Communication: “Nonlocal optimal design problems”

Link to the talk announcement

10. Annual meeting of the Unión Matemática Argentina (UMA). Venue: Facultad de Ciencias Exactas y Naturales, University of Buenos Aires, Buenos Aires, Argentina, December 2017. Communication presented: “Uniqueness of solutions for a semilinear problem involving the fractional Laplacian”, authors: Julián Fernández Bonder (University of Buenos Aires), Analía Silva (Universidad Nacional de San Luis), Juan Spedaletti (Universidad Nacional de San Luis).

Certificate link

11. VII Congress on Applied, Computational and Industrial Mathematics. Venue: Departamento de Matemática, Universidad Nacional de Río Cuarto, Río Cuarto, Argentina, 8-10 May 2019. Communication presented: “A constrained Shape Optimization Problem in Orlicz-Sobolev Spaces”, authors: João Vitor da Silva (University of Buenos Aires), Ariel Salort (University of Buenos Aires), Analía Silva (Universidad Nacional de San Luis) and Juan Spedaletti (Universidad Nacional de San Luis).

Certificate link

12. Annual meeting of the Unión Matemática Argentina (UMA), virtUMA2020 (online). Communication: “Continuity of solutions for the Δ_ϕ Laplacian operator”. September 2020.

Link to the talk

13. Departamento de Matemática, Facultad de Ciencias Exactas Físico-Químicas y Naturales, Universidad Nacional de Río Cuarto (UNRC), October 2020. Talk: “Asymptotic behaviour of eigenvalues for nonlocal problems”. Online.
14. X Congress on Applied, Computational and Industrial Mathematics, May 2025. Talk: “Ljusternik-Schnirelmann eigenvalues for the fractional m -Laplacian without the Δ_2 -condition”.

Refereeing for journals

1. Referee for the article “Sobolev spaces on canonical Banach spaces and Fourier transformations” for the Revista de la Unión Matemática Argentina. April 2023.
2. Referee for the article “Renormalized solutions for some degenerate elliptic problems in non-reflexive Musielak-Orlicz spaces” for the journal Proyecciones. December 2025-present.

Outreach activities

Event organization

Activity guide at the AniMATE San Luis 2012 outreach day, October 2012.

Certificate link

University service and governance

1. Alternate member of the governing council of the Facultad de Ciencias Físico Matemáticas y Naturales (FCFMyN), Universidad Nacional de San Luis (UNSL), representing the Departamento de Matemática, graduate constituency. (2016-present)

Link to the minutes in which I appear

2. Member of the academic affairs committee of the Facultad de Ciencias Físico Matemáticas y Naturales (FCFMyN), Universidad Nacional de San Luis (UNSL). (2016)
3. Member of the electoral board for the Departamento de Matemática, Facultad de Ciencias Físico Matemáticas y Naturales, Universidad Nacional de San Luis. (2019)

Resolution link RD-10-19/19 UNSL

4. Presiding officer at the 2019 UNSL elections for the faculty constituency of the Departamento de Matemática, Universidad Nacional de San Luis.

Link to the electoral board e-mail

5. Representative of the Departamento de Matemática before the Facultad de Ciencias Físico Matemáticas y Naturales for the institutional self-assessment process, 2021.

Resolution link RR-1-380/21 UNSL

6. Member of the electoral board for the 2025 UNSL elections, representing the Departamento de Matemática, Universidad Nacional de San Luis.

Resolution link RCD-3-231-24

7. Chair of the committee of the Licenciatura in Mathematics degree programme, Departamento de Matemática, Universidad Nacional de San Luis, 2026-present.

Member of faculty selection committees

1. Committee member for a first-category teaching assistant position, part-time, tenured. Departamento de Matemática, Universidad Nacional de San Luis. (2017)

Resolution link RCD-3-18/17

2. Committee member for second-category teaching assistant positions. Departamento de Matemática, Universidad Nacional de San Luis. (2017, 2019, 2021 and 2022)

Resolution link RDMat-3-60/17

Resolution link RDMat-3-156/19

Resolution link RDMat-3-13/21

Resolution link RDMat-3-75/22

3. Committee member for first-category teaching assistant positions, part-time, interim. Departamento de Matemática, Universidad Nacional de San Luis. (2020)

Resolution link RD-3-334/20

Resolution link RD-3-333/20

4. Committee member for first-category teaching assistant positions, part-time, tenured. Departamento de Matemática, Universidad Nacional de San Luis. (2017 and 2022)

Resolution link RCD-3-116/17

Resolution link RCD-3-95/22

Thesis committees

Licenciatura thesis committees

Committee member for the Licenciatura thesis in Mathematics of Natalí Ailín Cantizano. Departamento de Matemática, Universidad Nacional de San Luis. March 2017.

Link to thesis defence record

Master's thesis committees

1. Committee member for the Master's thesis in Mathematics of Luciano Luque. Departamento de Matemática, Universidad Nacional de San Luis. February 2020.

Resolution link RD-3-866/19 UNSL

2. Committee member for the Master's thesis in Mathematics of Paulo Guzman. Departamento de Matemática, Universidad Nacional de San Luis. November 2020.

Resolution link RD-3-266/20 UNSL

3. Committee member for the Master's thesis in Mathematics of Leopoldo Buri. Departamento de Matemática, Universidad Nacional de Río Cuarto. November 2020.

Resolution link M21120 UNRC

4. Committee member for the Master's thesis in Mathematics of Gustavo Abel Dorrego. Departamento de Matemática, Universidad Nacional de San Luis. April 2021.

Resolution link RD-3-410/20 UNSL

5. Committee member for the Master's thesis in Mathematics of María Laura de Borbón. Departamento de Matemática, Universidad Nacional de San Luis. October 2021.

Resolution link RD-3-623/21 UNSL

Ph.D. thesis committees

1. Committee member for the Ph.D. thesis in Mathematics of Luciano Luque. Departamento de Matemática, Universidad Nacional de San Luis. March 2020.

Resolution link RD-3-90/20 UNSL

2. Committee member for the Ph.D. thesis in Mathematics of Gustavo Abel Dorrego. Departamento de Matemática, Universidad Nacional de San Luis. April 2021.

Resolution link RD-3-186/21 UNSL

3. Committee member for the Ph.D. thesis in Mathematics of María Laura De Borbón. Departamento de Matemática, Universidad Nacional de San Luis. July 2022.

Resolution link RD-3-459/22 UNSL

4. Committee member for the Ph.D. thesis in Mathematics of Juan Federico Ramos Valverde. Departamento de Matemática, Universidad Nacional de San Luis. April 2025.

Resolution link RD-3-97/2025 UNSL

Other

1. Proficient in LaTeX on any editor (OS X/Mac, Windows and Linux).
2. Proficient in mathematical software:
 - Derive
 - Matlab
 - Maple
 - Octave
3. Excel and Office for OS X/Mac (Apple) and Windows.
4. Member of the Unión Matemática Argentina (UMA).
5. Member of the Asociación Argentina de Matemática Aplicada, Computacional e Industrial (ASAMACI).

Languages

1. English (English I and II of the undergraduate programme, and Instituto Target).
2. Italian (5 years of Italian at secondary school).